MPU5 Fabric Deep-Dive - 5-Switch Session (Fillable)

One-session sweep of the VLAN 17 (WAVES) fabric to settle packet paths, STP/root, ARP, and L2 loops. KEY STRUCTURAL FINDING from the configs: the Annex II Roof switch has TWO VLAN-17 uplinks - Fiber Gi0/1 to ALT_EOC 9300 (172.17.9.3) AND Cambium Fa0/5 to BLDG 602 (172.17.9.76) - both 'spanning-tree guard none'. Plus the WaveRelay mesh bridges VLAN 17 a third way. That is a real physical ring; RSTP must block a path. THREE switches also tie for VLAN-17 root at priority 24576. Pair this with the collection scripts (02 IOS-15, 03 IOS-XE, 04 path-trace). All read-only.

FABRIC MAP + SESSION TARGETS

Switch	Mgmt IP (VLAN17)	OS / script	Role on VLAN17
Annex II Roof	via 172.31 mgmt	IOS 15.2 / 02	Hosts .7 (Fa0/1). Dual uplink: Gi0/1 fiber + Fa0/5 cambium
ALT_EOC 9300 (AnnexII 3rd Flr)	172.17.9.3	IOS-XE 16.9 / 03	Far end of Annex FIBER (Te1/1/1 'to CCN-3'). root prio 24576
BLDG 602 9300	172.17.9.76	IOS-XE 16.9 / 03	Far end of Annex CAMBIUM (Gi1/0/4). Mesh enters Gi1/0/5. root prio 2457
EOC / RSOTOC SNW	172.17.9.2	IOS-XE 16.9 / 03	Core. root prio 24576 (3-way tie)
Chancery 2960	via 172.31 mgmt	IOS 15.2 / 02	Hosts .5 (known-good control MPU5)
RES 4 Relay	172.17.9.175	IOS 15.2 / 02	UK/British relay MPU5 on Fa0/4+Fa0/8

The ring: Annex Roof -> [Fiber] -> ALT_EOC 9300 -> core -> BLDG 602 -> [Cambium] -> Annex Roof. VLAN 17 rides all of it AND the MPU5 mesh. This is exactly where a loop or a flapping STP root would live.

A. SPANNING TREE + ROOT - PER SWITCH (FILL FROM SCRIPT 02/03 OUTPUT)

From each switch's 'show spanning-tree vlan 17' + summary. The question: is there ONE stable root, are the right ports blocking, and is anything reconverging?

Switch	Root bridge ID (who is root)	This sw root port	Blocked port(s) on V17	V17 TCN count
Annex Roof				
ALT_EOC 9300				
BLDG 602 9300				
EOC SNW				
Chancery				
RES 4 Relay				

Then 'show spanning-tree blockedports' + 'inconsistentports' on each:

Any inconsistent/guard ports (which sw/port):

Total switches reporting a DIFFERENT root for V17:

Any V17 TCN counter rising on re-check (which sw):

Decision: all switches must agree on ONE root for VLAN 17. If two disagree, or a TCN counter climbs over a few minutes, STP is unstable on the ring and that alone can cause intermittent reachability. The blocked port is the leg RSTP chose to break the ring - confirm exactly one V17 path is blocked, not zero (loop) and not flapping.

B. LOOP & DUPLICATE DETECTION (THE DIRECT LOOP TEST)

From script 04 + the MAC-move logs. A single MAC appearing on TWO ports of one switch = active L2 loop. A single IP with two MACs = duplicate IP.

.7 MAC (2290) - one port or many? (per sw):

Any MAC on 2+ ports of same switch? (Y/N+which):

Any 'MACFLAP/SW_MATM' log hits this session? (which sw):

Any duplicate-IP / DUP log hits? (which IP):

.135 MAC present ANYWHERE on V17? (Y/N):

This is the clean loop test across the whole fabric in one pass. We checked single switches before and found no flaps; doing all five at once catches a loop that only shows from a switch we hadn't queried.

C. ARP CENSUS ACROSS THE FABRIC (WHO IS REACHABLE, FROM WHERE)

'show ip arp | include 172.17.9' on each switch. Mark which nodes resolve from which switch. A node alive from one switch but not another tells us the path.

Node IP	Annex Roof	ALT_EOC .3	BLDG602 .76	EOC .2
.7 (172.17.9.7)				
.135 (BLDG602 MPU5)				

.5 (Chancery ctrl)				
.23 (HPSA7)				
Total V17 ARPs				

Write Y (resolves) / N (no entry) / I (incomplete) in each cell. Pattern matters: if .7 resolves from BLDG602 but not from ALT_EOC, the working path is via the cambium/mesh, not the fiber.

D. PACKET PATH / TRACEROUTE (HOW FRAMES ACTUALLY TRAVEL)

From script 04 (traceroute + ping bursts). Same-subnet so hops are few - the signal is whether it completes and via which next-hop, plus the up-window catch for .7.

traceroute .7 result (hops / unreachable):

traceroute .135 result:

traceroute .5 result (control):

.7 ping burst from BLDG602 (x/200):

.7 ping burst from ALT_EOC (x/200):

.135 ping burst (x/100, expect 0):

Running the .7 burst from BOTH the fiber-side switch (ALT_EOC) and the cambium-side switch (BLDG602) tells us which physical path carries .7's 7%. That narrows the fault to one leg.

E. TRUNK & UPLINK HEALTH (DIRTY LINK CAN DESYNC STP)

'show interfaces trunk' + 'show interfaces counters errors' on each. Focus on the ring legs: Annex Gi0/1 (fiber), Annex Fa0/5 (cambium), ALT_EOC Te1/1/1, BLDG602 Gi1/0/4.

Link	Status (up/down)	Errors/CRC?	STP state (FWD/BLK)
Annex Gi0/1 fiber			
Annex Fa0/5 cambium			
ALT_EOC Te1/1/1			
BLDG602 Gi1/0/4			
BLDG602 Gi1/0/5 mesh			

A trunk that is up but throwing CRC/errors can drop BPDUs intermittently, which makes STP reconverge - that would show as the rising TCN in Section A and explain flapping reachability without a hard link-down.

F. FINDINGS / SURPRISES

G. EDUCATED-DETERMINATION MATRIX

If the session shows...	Determination	Direction
2+ switches claim different V17 root, or TCN rising	STP unstable on the ring	Fix root priority / check BPDUs on ring legs
A MAC on two ports of one switch	Active L2 loop on VLAN 17	Find/break the doubled path (guard none is the risk)
One IP, two MACs (DUP log)	Duplicate IP on mesh	Reassign the duplicate (remote)
.7 resolves/pings via cambium but NOT fiber	Annex fiber leg or its STP state is the issue	Inspect Gi0/1 fiber + ALT_EOC end
All STP stable, no loop, no dup, .7 still 7%	Fabric is healthy; .7 is a NODE/RF problem	Back to the .7 node worksheet (bridging/RF)
.135 absent from all ARP/MAC fabric-wide	Confirms .135 off the network entirely	Physical/console (as expected)

H. HOW TO RUN THIS SESSION (ORDER + SAFETY)

1. Enable MobaXterm session logging on EACH tab before pasting (one log file per switch).
2. IOS-15 switches (Annex Roof, Chancery, RES 4 Relay): paste **02-fabric-collect-IOS15.ios**.
3. IOS-XE switches (BLDG602, ALT_EOC, EOC): paste **03-fabric-collect-IOSXE16.ios**.

4. On BLDG602 AND ALT_EOC, also paste **04-packet-path-trace.ios** (the .7 dual-path burst).
5. Save all logs. Transcribe the key cells above; attach logs under Section F.

   SAFETY: every command is show/ping/traceroute. No config mode, no write memory, no clear, no shutdown. The ping/traceroute bursts generate light traffic only. Nothing changes STP or the fabric.

HONEST READ

This session is built to DECIDE between two worlds: (1) the fabric itself is unstable on the Annex VLAN-17 ring - a flapping STP root, a loop hidden behind 'guard none', or a dirty trunk dropping BPDUs - which would be a network-side cause consistent with your read; or (2) the fabric is clean and .7's problem is truly at the node/RF level. The dual-uplink ring + 3-way root tie are real config facts that justify the deep dive - but they are not yet proof of a fault. Earlier single-switch checks showed no MAC flaps; this widens the net to all five at once. I am not asserting a loop exists - I am giving you the one-pass data to confirm or rule it out.